Diploma in Data Analytics

Multiple linear regression continued

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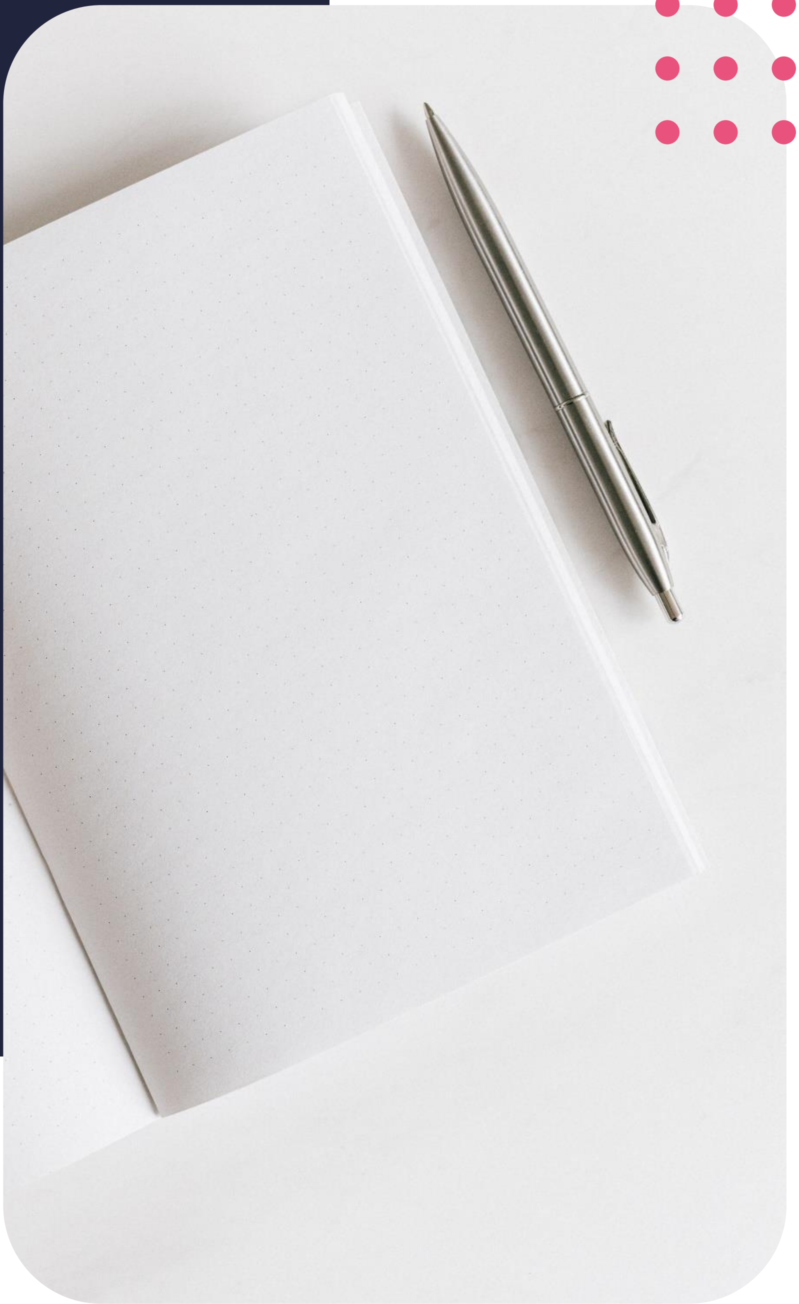


Lesson outcomes

By the end of this lesson, you should be able to:

- Know when to use the multiple linear regression model
- Know if the multiple linear regression model is a good fit for the data
- Know the basic concepts surrounding logistic regression

Introduction

In this lesson, we will elaborate on the principles of multiple linear regression. We will elaborate on the assumptions that accompany multiple linear regression, how to simplify a multiple linear regression model, and problems that can occur when fitting a multiple linear regression model to the data.
Thereafter, we will discuss what happens if your model does not fit a linear trend well, in other words, if the data is non-linear.

The lesson will end by introducing logistic regression.

Multiple linear regression continued

Multiple linear regression recap

In the previous lesson, we learnt that simple linear or ordinary linear regression is a function that compares the dependent or outcome variable against one independent or explanatory variable in order to make predictions. We also learned that in practise, it is uncommon to have a dependent variable explained by one independent variable and more likely for the outcome variable to be explained by more than one independent variable. This is called multiple linear regression.

Uses of linear regression

Linear regression is a popular method in statistical analysis of data. Main uses for this method include:

- The ability to forecast the result of an event. In other words, multiple linear regression can tell us how much the chance of death is expected to increase or decrease for every one-point increase in the person's age.
- Secondly, multiple linear regression's use is to be able to recognize the strength of the effect the independent variable has on the dependent variable. It can tell us more about the strength of the relationship between the variables.
- Another major use for multiple linear regression is the forecasting of trends. To be able to determine what the price of a specific car will be in a year's time from now is one of the uses of multiple linear regression.

Assumptions of linear regression model

However, when we apply a multiple linear regression model to the data, there are certain assumptions that we assume to be true:

- Some of these assumptions include that there exists a linear relationship between the dependent and independent variables.

- Secondly, we assume that the correlations between the independent variables are not too high, meaning that multicollinearity does not exist between the variables.
- Thirdly, that all independent variables are randomly and independently sampled from the population. Therefore, we assume that all independent variables are independent random.
- Lastly, we assume by using the multiple linear regression model that the residuals are normally distributed and have a mean of zero and variance of sigma.

Selection processes

When applying the multiple linear regression, the model can quickly include a lot of variables. But are all variables necessary to include in the model? Are all variables of equal importance to include in the model?

We have previously seen that we can use the f statistic and p values to determine some of the variable importance, but if our p-value is large, there can be incorrect conclusions drawn.

We can determine which predictor variables are associated with the response variable through variable selection. Variable selection means to select certain variables to include in the mode, in other words, including a subset of variables in the model.

Forward selection

Forward selection is one of the ways we can perform variable selection on the linear model. Forward selection involves starting out with a small model, for example the intercept, and adding a variable to the model with each step. At each step of the expansion process of the model, the model is tested by criterion to judge the quality of the model. This criterion could be the lowest p-value, the highest adjusted R^2 value, the lowest Mallow's Cp or the lowest AIC value, to name a few. When the criterion stops improving, we have found our model.

Backward selection

Backward selection starts with all the variables in the model and at each step it removes the variable with the highest p-value from the model. In other words, with each step of the backwards selection process, the variable that has the highest p-value is removed from the model. We would choose a certain criterion to stop at beforehand, for instance, we could stop the selection process when the variables in the model are all below a certain p-value.

Mixed selection

The last selection process we will discuss today, is called mixed selection.

Mixed selection is a combination of the forward and backward selection processes we just learned about.

The Mixed selection process starts off with no variables in the model and adds variables with each step, just like the forward selection process would. We would carry on with the forwards and backwards selection process until all the variables in the model p-values are adequately low.

The reason why we would consider the mixed selection process over solely the forwards selection process, is because forwards selection can include variables in the model that can later become redundant. The mixed selection process also provides a smaller model than through the backwards selection process.

Model fit statistics

So how do we now test which of these models are the best fit for the data? There are various model fit statistics we can use to determine if the model fits the data well:

- R^2, the coefficient of determine
    - Used to measure the variation in the outcome variable. We also know that as the variables we have in the model increases, the R^2 statistic also improves. Therefore, we need to use other statistics as well to determine the model fit
- Residual standard error or RSE statistic.
    - We know that the residual standard error is a measure of how well the linear regression model fits the data.
- Mallow's Cp
- Akaike information criterion (AIC)
- Bayesian information criterion (BIC),
- Adjusted Akaike information criterion
- Adjusted Bayesian information criterion
- Adjusted R^2

All of these are fit statistics that can be used to determine the goodness of the model fit, in other words, to determine if the given distribution fits the model well.

Problems

Thus far we have just discussed fitting a linear model to a specific given data set. A linear regression model comes with a given set of assumptions that should be met if we choose to fit this model to the data. If assumptions are not met, we need to be on the lookout for problems that can jeopardize the quality of the results.

Some of these problems include:

1. Non-linearity of the response-predictor relationships

2. Correlation of error terms

3. Non-constant variance of error terms

4. Outliers

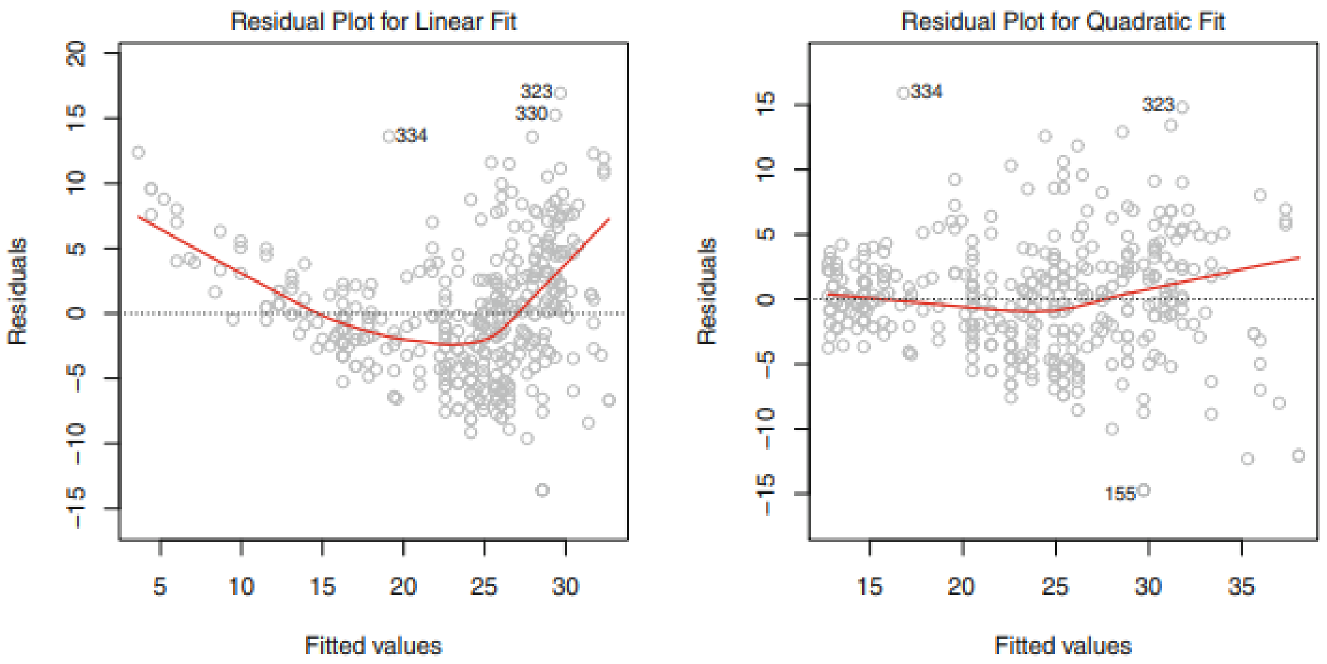
5. High-leverage points

6. Collinearity

Non-linearity

We know that the linear regression model assumes that there is a linear relationship between the independent and dependent variables, but if the relationship between those variables are not linearly distributed and we draw all the conclusions as if it is, the accuracy of the model and results drawn will be severely flawed.

We can create residual plots to help us identify non-linearity in the model. The ideal is to see a residuals plot that shows no noticeable pattern between the independent and dependent variables. If there is a noticeable pattern, it indicates that there might be a complication with our chosen model, and it might not be the best fit for the data.



(James et al., 2013)

If we look at the plot comparison for the Auto dataset that represents the miles per gallon to horsepower of a car, the left-hand plot is the representation of a linear regression where a pattern is visible. This indicates that the data might be non-linear. The right-hand side plot is a quadratic fit of the same dataset. In this plot a less discernible pattern is visible, indicating that this model might be a better fit for the data than the linear regression model is.

Correlation of error terms

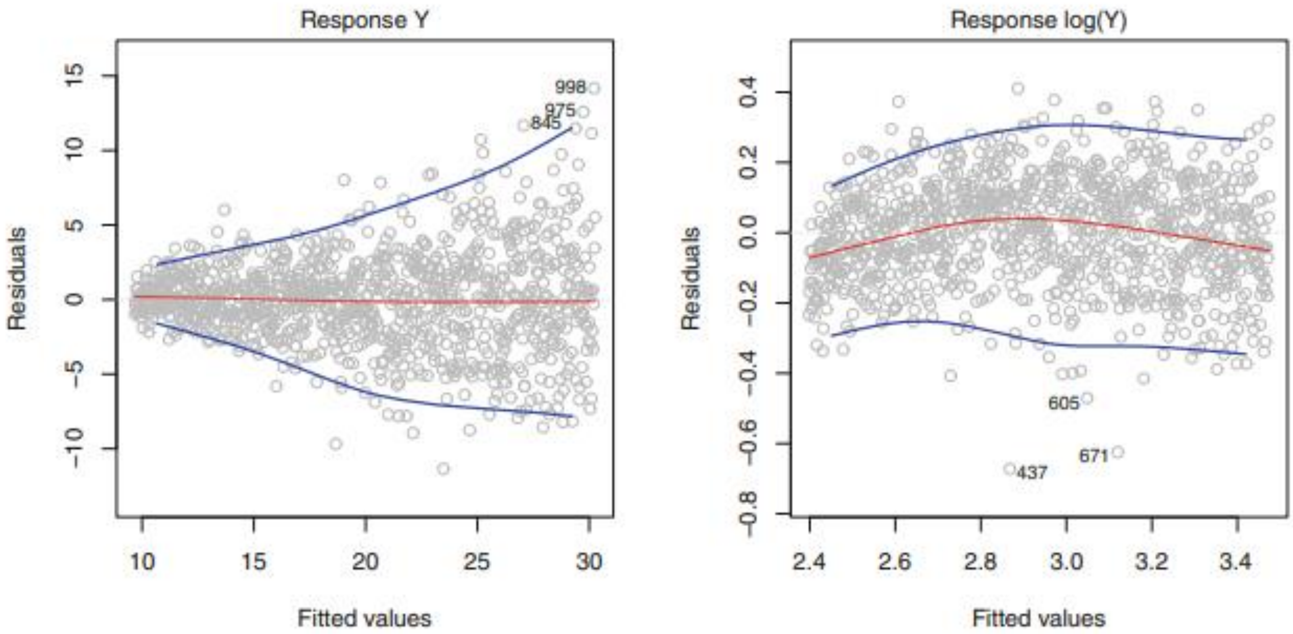
Another assumption that is made when using the linear regression model, is that the error terms, for example e_i and e_j that are associated with the i^{th} and j^{th} observations, are uncorrelated. If there is correlation between the error terms, it indicates that there is additional intelligence in the data not yet utilized by the current model used.

Correlation between the error terms, can be the justification for the estimated standard errors to underestimate the true standard errors leading to the confidence interval bounds being smaller than they ought to be.

Non constant variance of error terms

The third assumption we make when we fit a linear regression model to the data, is that the error terms have a constant variance. This means that the error terms have a mean of zero.

We can check if the variance is constant through a variety of tests but looking at a visual representation in the form of a residuals versus fitted values plot is generally adequate.



(James et al., 2013)

If we look at the plots that show the residual values versus the fitted or predicted values, the ideal is not to see a pattern. The errors should seem random with no increasing or decreasing pattern visible.

The left-hand side plot shows an increase of the residuals with the fitted values. This plot indicates that the variance is non-constant. The right-hand side plot is a log transformation of the left-hand side plot. The transformation appears to have helped the residuals to now have a constant variance, although there does appear to be some non-linear relationship.

Outliers

- An outlier is a point that is far off from the model's value that is predicted.
- An outlier varies significantly from other data points.
- We can use a residual plot to spot outliers in the data.
- Outliers can occur for a variety of reasons, including a data point being recorded incorrectly or they can occur because of natural causes, like a novelty data point.

High leverage points

High leverage points are data points that have high leverage if they have extreme predictor independent variables. When dealing with multiple linear regression, multiple predictors can have extreme independent values that have extremely high or low values for multiple predictors.

The difference between a high leverage point and an outlier, is that we regard an observation to be an outlier if and only if it has an extreme value with regards to other y-values, not other x-values.

Collinearity

The last potential problem that can be encountered when dealing with multiple linear regression we will discuss in this lesson, is collinearity. Collinearity is when two or more independent variables are closely related to each other. Collinearity will likely increase the variance of at least one estimated regression coefficient which can cause regression coefficients to have the wrong sign. We are not going to discuss ways to remedy these problems in this lesson, I merely want you to be aware of problems that can arise and not blindly apply the multiple linear regression model to each given set of data, but to look further .

Logistic regression

Why not linear

We have previously discussed all the problems that can occur when applying the linear regression model to the data, but what happens when the response variable is not quantitative, but qualitative? Is linear regression still the best model to fit the data?

Suppose we gather data to predict what marketing stream will produce the biggest leads for our campaign. We could look at social media streams like Facebook, Instagram and Twitter and encode these values as a quantitative variable with 1, 2 and 3.

$$Y = \begin{cases} 1\ if\ Facebook; \\ 2\ if\ Instagram; \\ 3\ if\ Twitter. \end{cases}$$

However, there is no quantifiable difference between these variables. We can choose a different type of encoding for the variables, and it would be equally reasonable. But applying a linear model to these different encodings would provide radically different results. We can see that when there are more than 2 qualitative responses, a linear model is not the best option.

Logistic regression defined

- Binary logistic regression predicts if something is true or false, did the passengers survive or not?
- Unlike linear regression, logistic regression fits an s-shaped curve to the data that spans from 0 to 1. The curve therefore shows us the probability of the passengers surviving based on the predictor variables we included in the model.
- Logistic regression is usually used for classification, so the model will classify the survival rate of the passenger into the category of survived or not survived.

Logistic regression in R

We use the glm() function in R to fit a logistic regression model to the data. If the outcome variable is binary, in other words if like in the case of the Titanic dataset the passengers could either survive or not, we have to specify to R that the family of the model is binary.

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